



Module-4: First Law of Thermodynamics

Thermodynamics (NMEC103) : (Winter 2024-25)

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General Conservation Laws

Balance Equation

$$X_G = X_T + X_p$$
$$\dot{X}_G = \dot{X}_T + \dot{X}_p$$

A quantity is conserved if a quantity can neither be produced nor destroyed. Hence the balance equation for conserved quantities

Conservation Equation

$$X_G = X_T$$
$$\dot{X}_G = \dot{X}_T$$

So far scientists have **empirically discovered 4 major entities that are conserved**

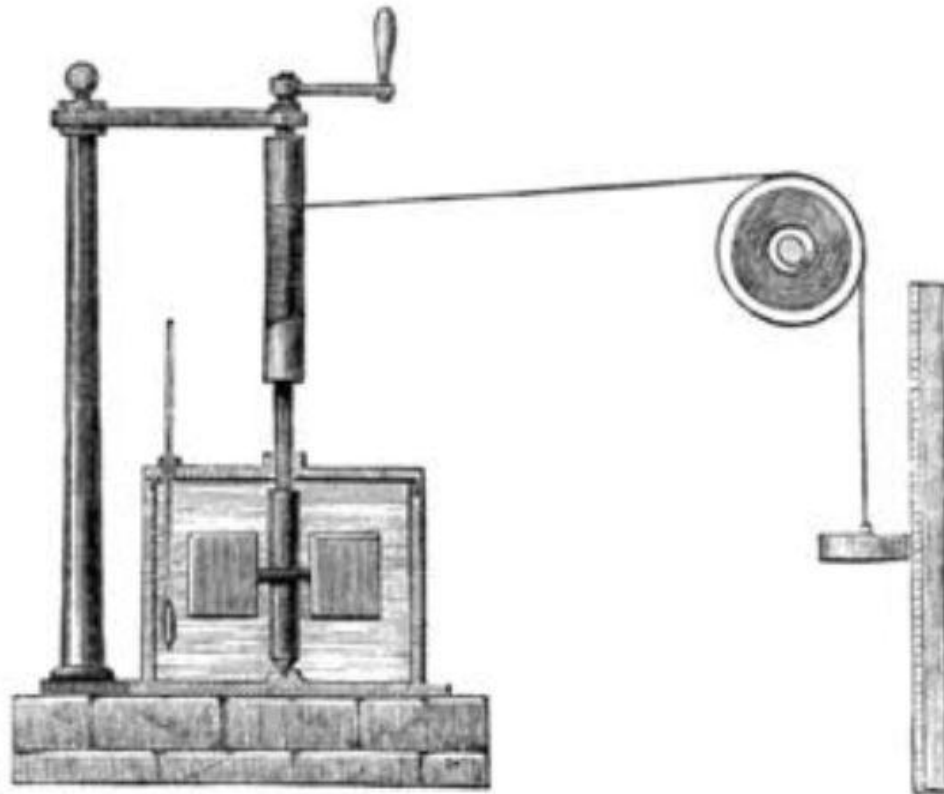
1. Mass (in non-nuclear reactions)
2. Momentum (both linear and angular)
3. Energy (Total)
4. Electrical Charge

Note: Entropy (generated) and Exergy (destroyed) are not conserved quantities

1st law of Thermodynamics

- Although heat and work interactions were known independently for a long time, a detailed understanding of their inter-relationship was not known until the mid-19th century.
- Also, it was not realized that heat is a form of energy interaction. The prevailing theory for heat in the early times was the caloric theory, which likened heat to an invisible fluid. It was thought that just as water flows from a higher level to lower level, heat flows from a higher temperature to a lower temperature.
- It was James Prescott Joule, who between the years 1840 to 1849, clearly showed the equivalence of heat and work through his experiments. He computed mechanical work W as the product of force into distance, and heat in terms of the temperature rise using the expression $Q = M.C.\Delta T$. He demonstrated that the magnitudes of work input and heat input required for achieving the same temperature rise in a substance always had a fixed ratio, $W/Q = 4.186 \text{ N.m/Calorie}$, which is known as the “Mechanical equivalent of heat”.

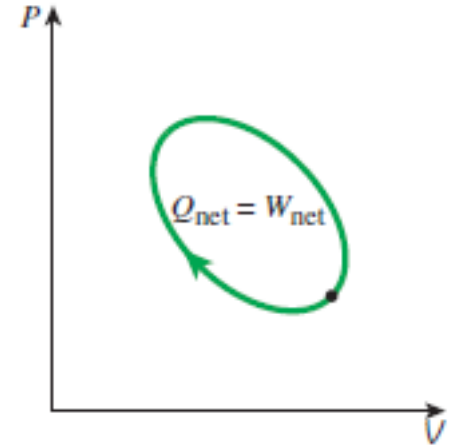
The set up used by Joule for arriving at this conclusion is shown in Fig. 1. It consists of a known mass of water whose temperature is raised by the operation of a paddle wheel. The work input for paddle wheel is calculated as torque T angular displacement θ ; the torque in turn, is given as the force due to the mass hung from the rope $\times$ radius of the shaft which drives the paddle wheel. Equivalent heat is calculated as $Q = \text{mass of water} \times \text{specific heat of water} \times \Delta T$ for water.



1st law of Thermodynamics

First Law is the law of conservation of energy. According to 1st law energy can neither be created nor be destroyed, it can only transform from one form to other.

First law talks about the quantitative conservation of total energy and is silent about the quality or grade of energy i.e. according to 1st law adding 5 kJ of heat or work to a system are equivalent. The total energy includes the Mechanical and Thermal, chemical, Nuclear Energies. Unlike the total energy the mechanical energy is not always conserved (Bernoulli Equation)

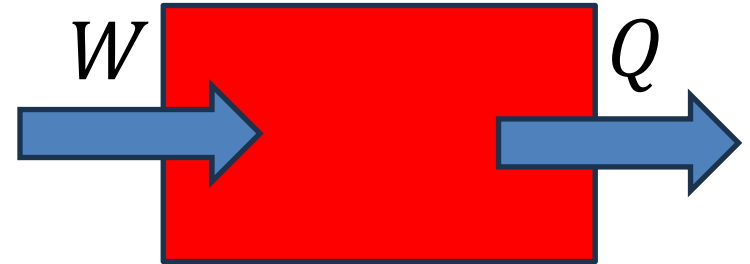
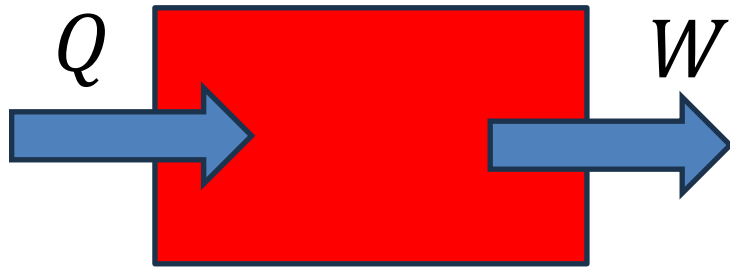


First law for a cycle: $\oint \delta W = J \oint \delta Q$

This is a constraint imposed by 1st law on a thermodynamic cycle

The mechanical equivalent of heat is just a proportionality constant. Its value depends on the unit in which heat and work are expressed $J=1 \text{ Joule/Joule}, 4.186 \text{ Joule /Calorie}.$

- The first law for a cyclic process can be understood as follows: Since the system returns to its initial state at the end of the cycle, there is no net storage of energy in the system. Therefore, the net heat and work interactions between the system and surroundings should balance (i.e. input = output) for the whole cycle, irrespective of whether their values are positive or negative

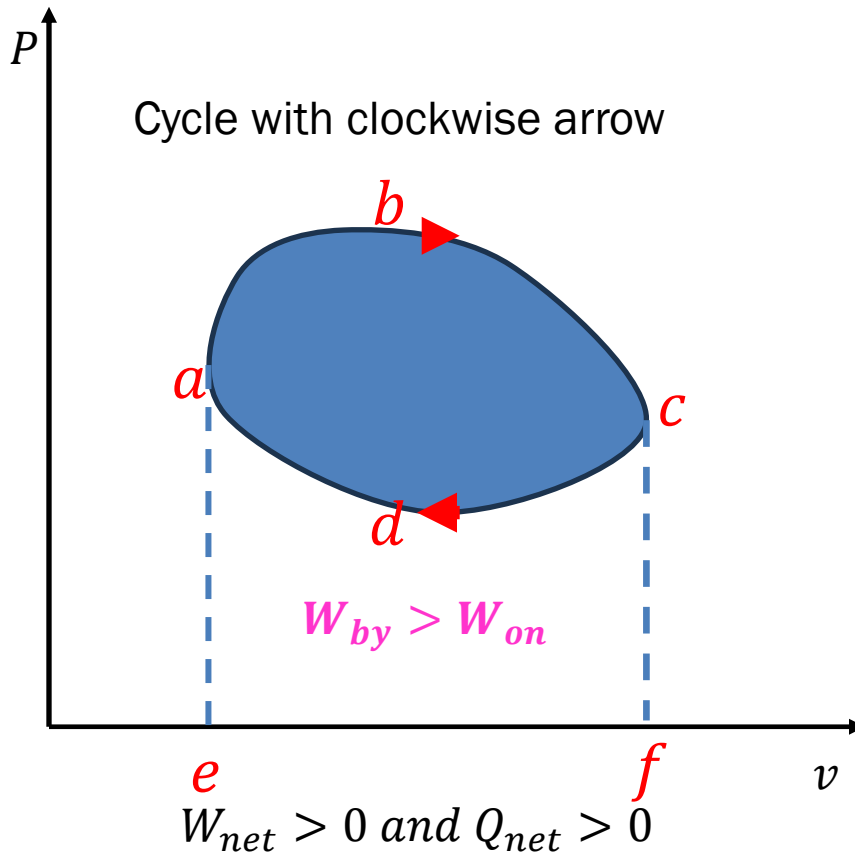


$$+ \quad W_{by} = \text{Area } e - a - b - c - f$$

$$- \quad W_{on} = \text{Area } f - c - d - a - e$$

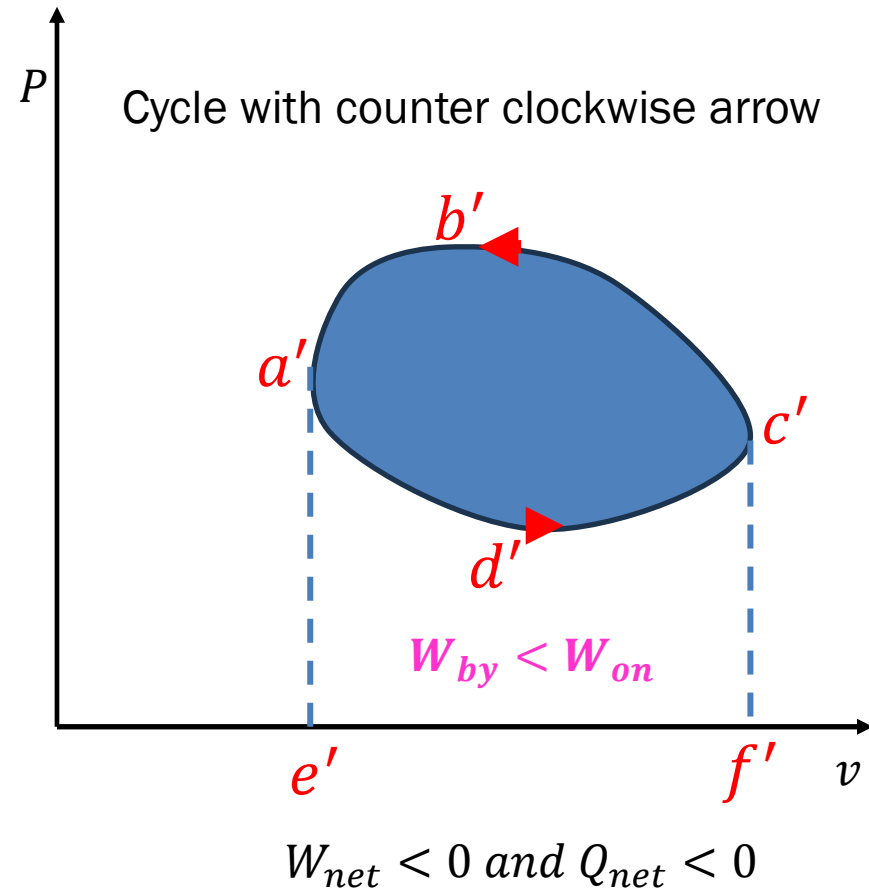
$$+ \quad W_{by} = \text{Area } e' - a' - d' - c' - f'$$

$$- \quad W_{on} = \text{Area } f' - c' - b' - a' - e'$$



Work producing machine

Heat Engine or Power Cycle



Work Consuming machine

Thermodynamic Device or Refrigeration Cycle

The 1st Law of Thermodynamics for a Process Undergone by a Closed System

- Consider a system that undergoes a cycle in which it changes from state 1 to state 2 by process A and returns from state 2 to state 1 by process B. This cycle is shown in Fig. 1 on a pressure (or other intensive property)–volume (or other extensive property) diagram. From the first law of thermodynamics

$$\oint \delta Q = \oint \delta W$$

$$\int_1^2 \delta Q_A + \int_2^1 \delta Q_B = \int_1^2 \delta W_A + \int_2^1 \delta W_B$$

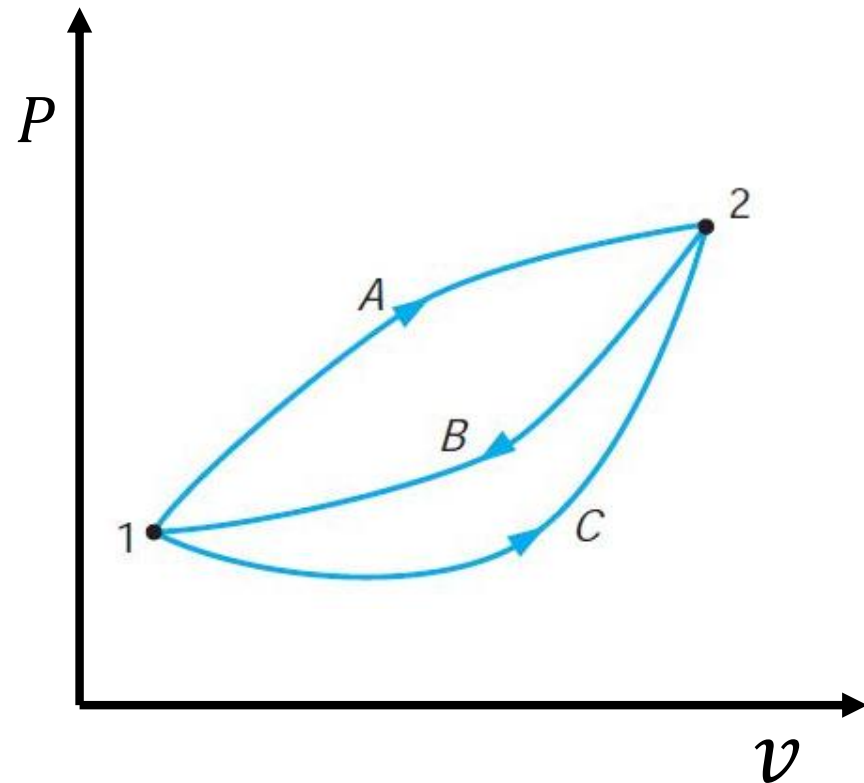
$$\int_1^2 \delta Q_C + \int_2^1 \delta Q_B = \int_1^2 \delta W_C + \int_2^1 \delta W_B$$

$$\int_1^2 \delta Q_A - \int_1^2 \delta Q_C = \int_1^2 \delta W_A - \int_1^2 \delta W_C$$

$$\int_1^2 \delta Q_A - \int_1^2 \delta W_A = \int_1^2 \delta Q_C - \int_1^2 \delta W_C$$

$$\int_1^2 (\delta Q - \delta W)_A = \int_1^2 (\delta Q - \delta W)_C$$

$$E_2 - E_1 = Q_{1-2} - W_{1-2}$$



First law when applied to a process undergone by a closed system a new property called energy E (Stored Energy) is introduced.

First law for a process for a closed system : $\frac{dE}{dt} = \dot{Q} - \dot{W}$

ΔE

$$= \Delta U + \Delta KE + \Delta PE + \Delta(\text{magnetic energy}) + \Delta(\text{electric energy}) \\ + \Delta(\text{surface energy}) + \Delta(\text{solid distortion energy})$$

First law for a process for a closed stationary simple system:

$$\frac{dU}{dt} = \dot{Q} - \dot{W} \\ \Rightarrow U_2 - U_1 = Q_{1-2} - W_{1-2} \\ \Rightarrow m(u_2 - u_1) = m(q_{1-2} - w_{1-2})$$

$\dot{W}$ is the rate at which the system does work on the surrounding through the surface forces. The contribution of the body forces is already accounted for by the inclusion of the potential energy term in the E and so is excluded from $\dot{W}$ to avoid duplication.

Where u is the internal energy which includes the Thermal energy (molecular kinetic & potential energy), chemical energy, nuclear energy and energy due to other force fields.

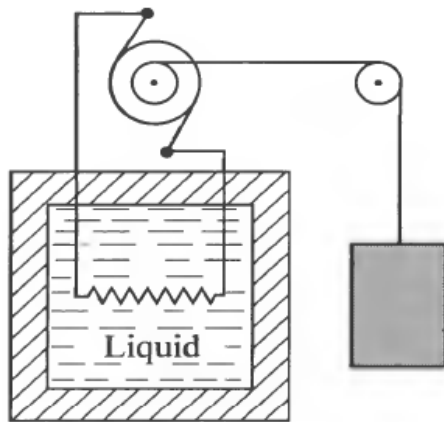
Caratheodory Axiom-I

If a closed system undergoes an **adiabatic change of state**, the work transfer experienced by the system is the same for all adiabatic paths connecting the same initial and final states i.e. the total value of work interaction causing the change of state depend only on the end states and not on the details of the process

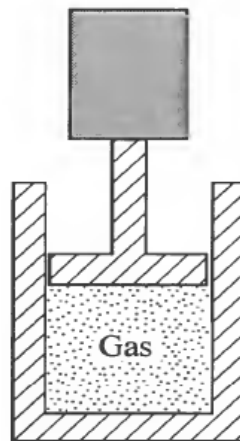
$$\delta W = -dE$$

Conservative field and E/U is the potential of that field

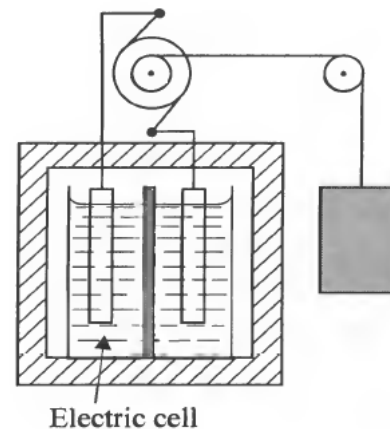
In this case work is used to define the change in energy E of a closed system thus



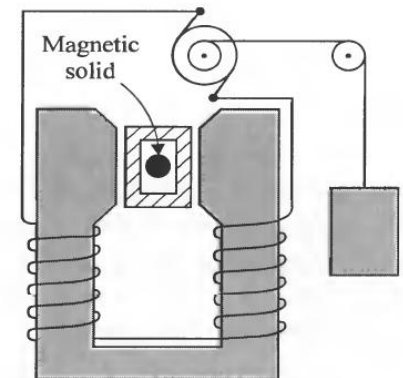
Adiabatic
electrical
work



Adiabatic
displacement
work

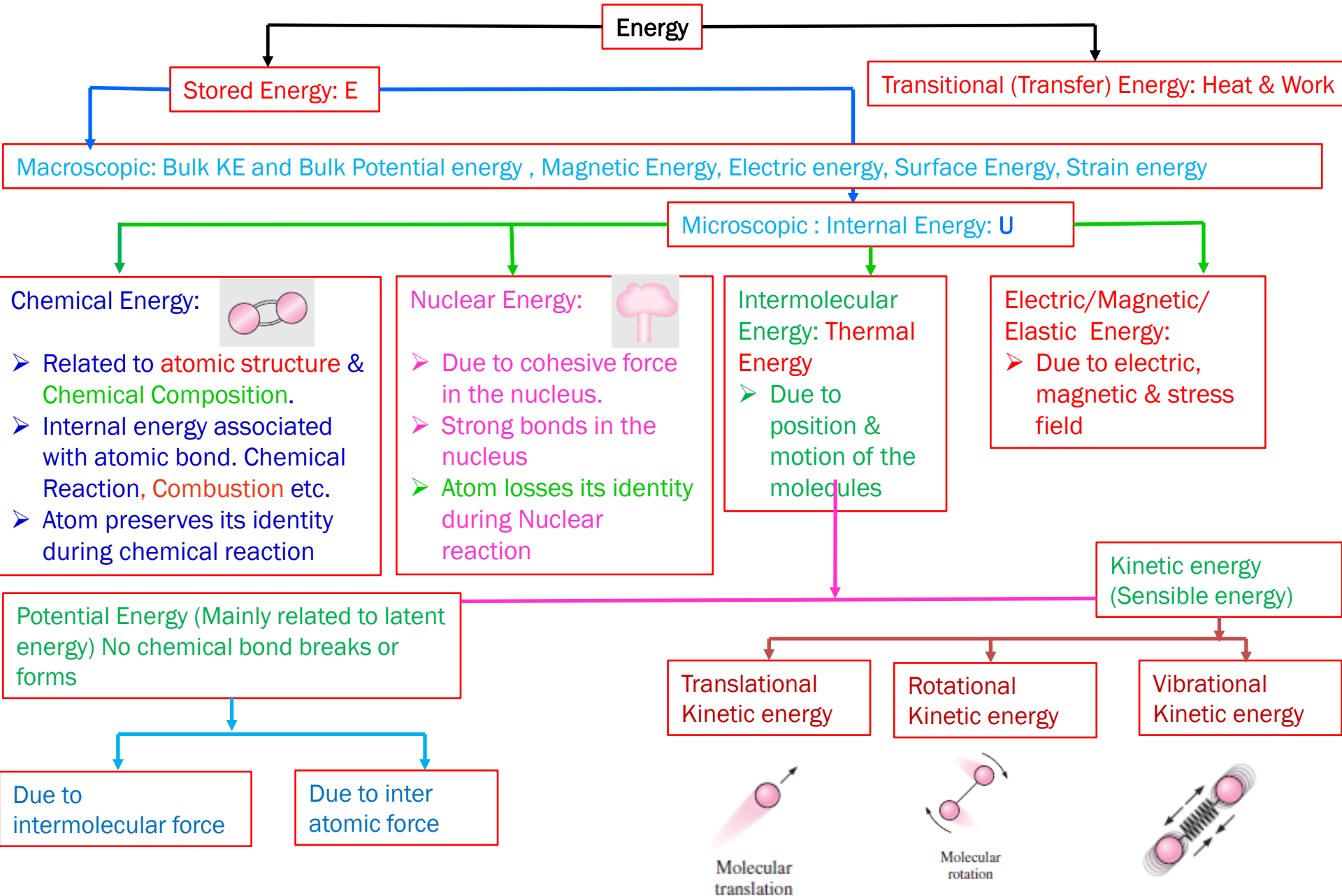


Adiabatic electro
chemical
work



Adiabatic
Magnetic
work

Broad Classification of Energy



Important Note

- ❖ Note: When the system is subjected to stress field, within the elastic limit the energy is stored as elastic potential energy (strain energy). But once the system is deformed beyond the elastic limits into the plastic energy the excess energy stored as internal energy not as macroscopic potential energy.
- ❖ Note: Similarly, when system is magnetized and demagnetized without any hysteresis then the energy will be stored as potential energy of the system

Kinetic Energy

Let us consider the body as it moves from $s = s_1$, where the magnitude of its velocity is $\mathbf{V}_1$, to $s = s_2$, where its velocity is $\mathbf{V}_2$. Assume for the present discussion that the only interaction between the body and its surroundings involves the force $\mathbf{F}$

$$F_s = ma = m \frac{d\mathbf{V}}{dt} = m \frac{ds}{dt} \frac{d\mathbf{V}}{ds} = m\mathbf{V} \frac{d\mathbf{V}}{ds}$$

$$\int_{V_1}^{V_2} m\mathbf{V} d\mathbf{V} = \int_{s_1}^{s_2} F_s ds;$$

$$\frac{1}{2} m(V_2^2 - V_1^2) = \int_{s_1}^{s_2} \mathbf{F} \cdot d\mathbf{s}$$

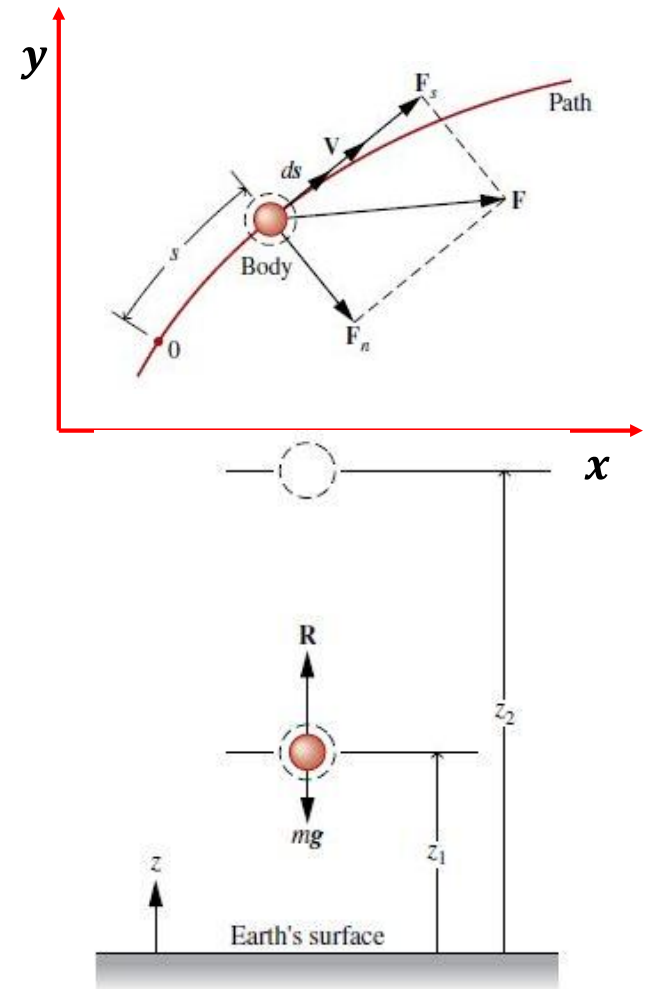
Potential Energy (No resistance)

$$\delta W = -Fdz = -dPE$$

$$F = ma = mg$$

$$dPE = Fdz = mgdz$$

$$PE_2 - PE_1 = mg(z_2 - z_1)$$



Different Forms of 1st law for a closed system

$$dE = dU + mVdV + mgdZ$$

$$E_2 - E_1 = \left(U_2 - U_1 + \frac{m}{2} (V_2^2 - V_1^2) + mg(Z_2 - Z_1) \right)$$

Differential form: $\delta Q = dE + \delta W$

Differential rate form: $\frac{\delta Q}{\delta t} = \frac{dE}{dt} + \frac{\delta W}{\delta t}$

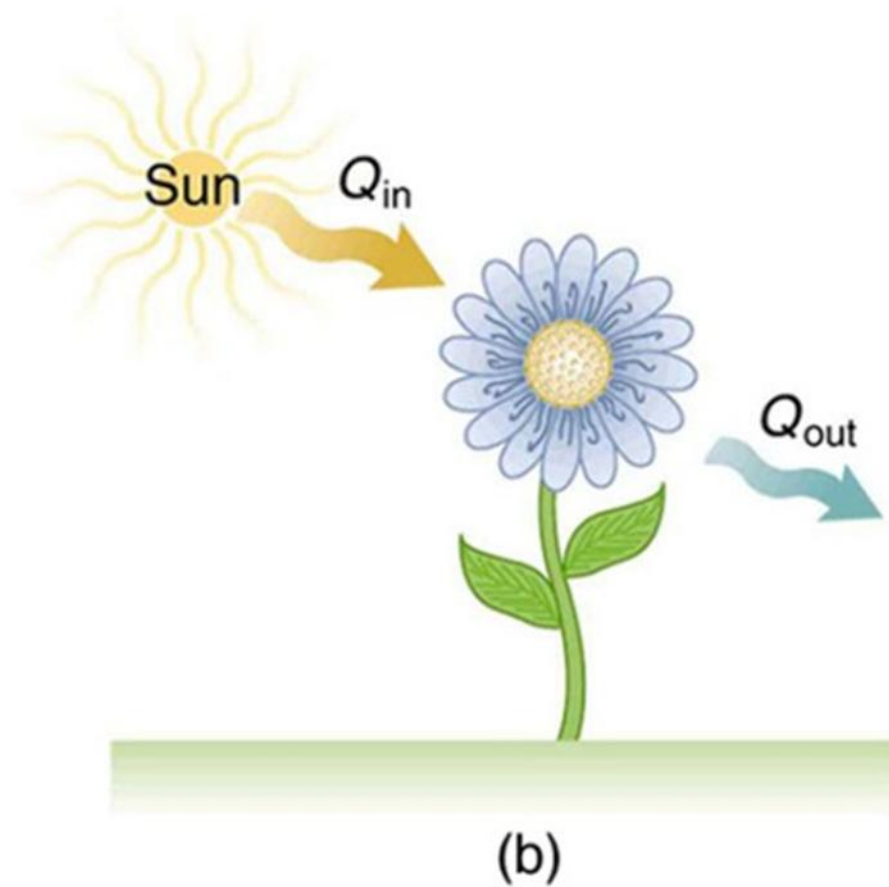
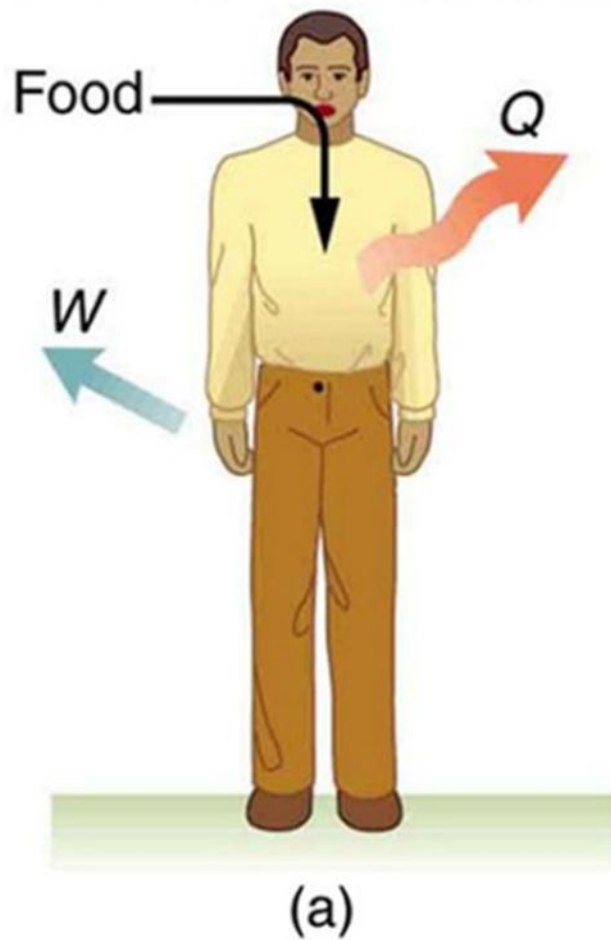
Integral form: $Q_{1-2} = E_2 - E_1 + W_{1-2}$

The rate equation form of the 1st law is

$$\dot{Q} = \dot{W} + \frac{dU}{dt} + \frac{d(KE)}{dt} + \frac{d(PE)}{dt}$$

$$Q_{1-2} = \left(U_2 - U_1 + \frac{m}{2} (V_2^2 - V_1^2) + mg(Z_2 - Z_1) \right) + W_{1-2}$$

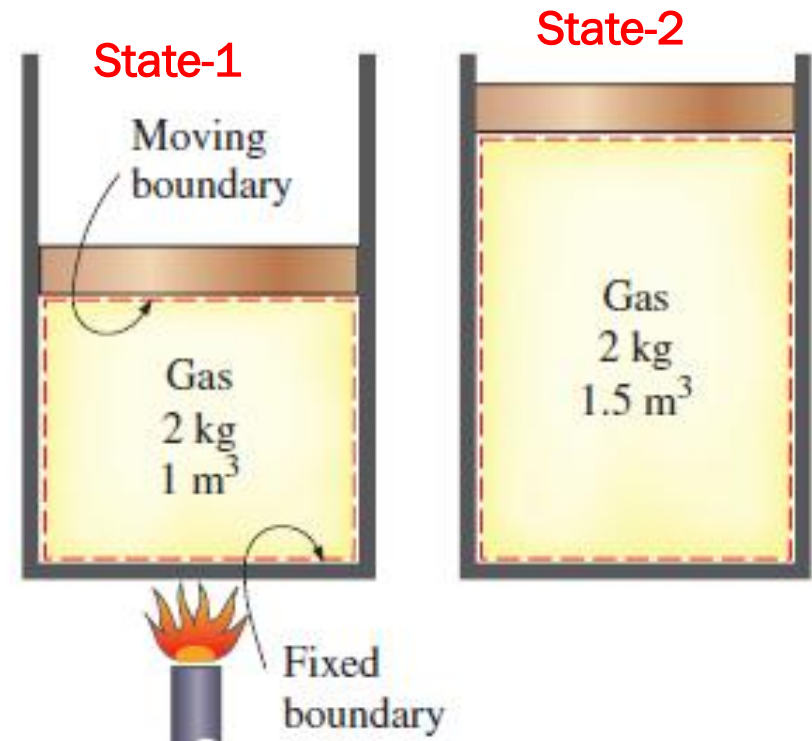
$$\Delta U = -Q - W + \text{food energy} \quad \Delta U = \text{stored food energy}$$



What is enthalpy (Greek words for heat inside) ?

Enthalpy is a **combined property** defined as: $H = U + pV$; $h = u + pv$

It is a **misconception** that enthalpy is **internal energy + flow work**. The **pv** term in the enthalpy not necessarily be the flow work, it can come from any source. In case of constant pressure heating of a closed system it can be shown that the heat interaction is equal to the change in enthalpy of the system (**also valid for a steady flow process**). In this case pv term is coming from the moving boundary work not from the flow work.

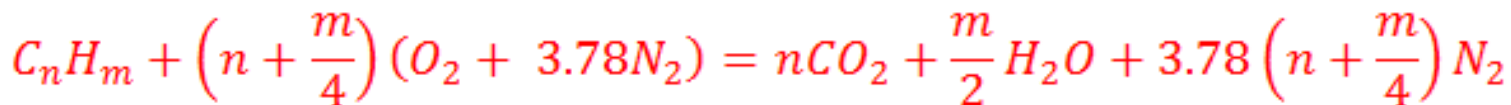
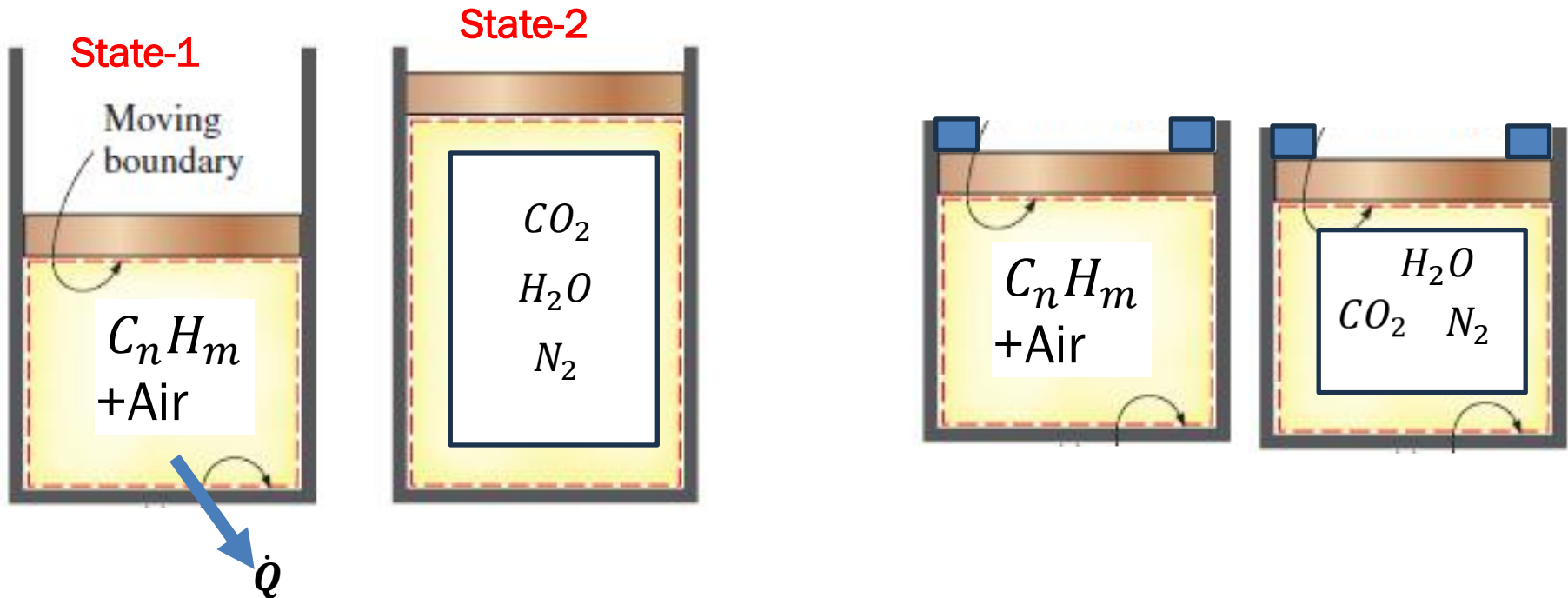


$$Q_{1-2} = U_2 - U_1 + W_{1-2} = U_2 - U_1 + \int P dV$$

$$Q_{1-2} = U_2 - U_1 + P(V_2 - V_1) = H_2 - H_1$$

It should be noted that **internal energy** represents a form of stored energy, but **pv does not**; therefore their sum i.e. **enthalpy is not a stored energy so enthalpy should never be defined as the energy content of the body**. The enthalpy is simply a useful property defined as a combination of other properties and is not a form of energy.

- ❑ Suppose we burn a certain amount of hydrocarbon fuel in a container fitted with a movable piston. As the fuel burns it produces carbon dioxide and water vapor, which occupy more space than original fuel and oxygen, and as a result the piston is driven out to accommodate the products.
- ❑ This expansion requires work, which is taken from the energy released in a combustion. If the combustion takes place in a container with rigid wall, the combustion release the same amount of energy, but none of it is used to do work because no expansion can occur. In other words, more energy is available as heat in the latter case than in the former.



- ❑ To calculate the heat that can be produced in the former case, we have to account for the energy that is used to make room for CO₂ and water vapor and subtract that from the total energy, this is true even if the fuel burns in a dish-because, although we can not see it so readily, the gaseous products must still make room for themselves.
- ❑ Thermodynamics have developed a clever way of taking into account the energy used to do work when any change, and particularly the combustion of the fuel, without having to calculate the work explicitly in each case. The energy released as heat by a system free to expand or contract as a process occurs is exactly equal to the change in enthalpy of the system
- ❑ As an actual exam the enthalpy change accompanies the combustion of 1 litre of gas is about 33 MJ. Therefore we know without any further calculation that burning a liter of gasoline in an open container will provide 33 MJ of heat. In the same combustion 130 kJ is used to make room for the gases that are generated.

Specific Heats: Specific heat is defined as the amount of energy required per unit mass to raise the temperature by one degree. It is assumed that the substance remain in single phase and undergo sensible heating/cooling process.

Simple Compressible Substance:

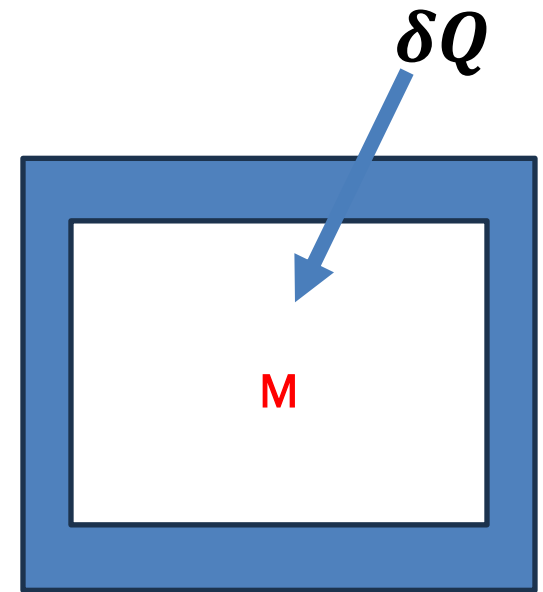
Consider a closed stationary system undergoing a quasi-static process

$$\delta Q = dU + \delta W$$

$$\delta Q = dU + p dV$$

The process is isochoric ($dV = 0$)

$$\delta Q = dU$$



If C_v is the specific heat for this constant volume process the

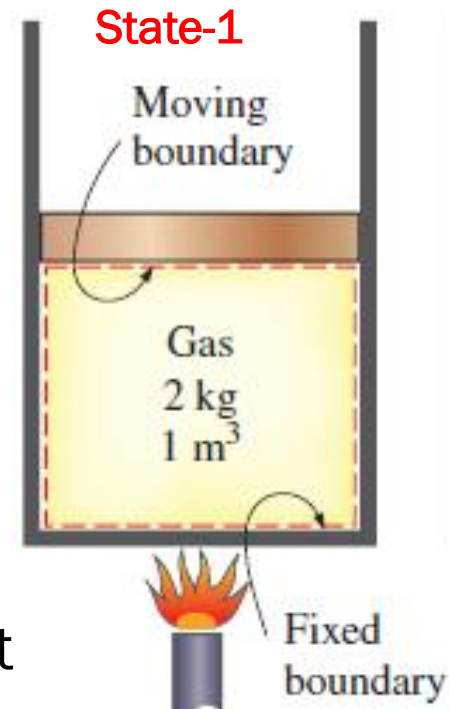
$$C_v = \frac{1}{M} \left(\frac{\delta Q}{\delta T} \right)_{\delta T \rightarrow 0} = \frac{1}{M} \left(\frac{\partial U}{\partial T} \right)_v = \left(\frac{\partial u}{\partial T} \right)_v \quad \delta Q \Rightarrow du = C_v dT$$

- i. If the process is **isobaric** ($P=\text{constant}$)

$$\delta Q = dU + pdV = dH$$

C_p , the specific heat at constant pressure =

$$C_p = \frac{1}{M} \left(\frac{\delta Q}{\delta T} \right)_{\delta T \rightarrow 0} = \frac{1}{M} \left(\frac{\partial H}{\partial T} \right)_p = \left(\frac{\partial h}{\partial T} \right)_p$$



Solids & Liquids: Both the phases are nearly constant density substances

$$dh = d(u + pv) = du + pdv + vdp \approx du + vdp$$

For both solids & liquids v is very small $\Rightarrow vdp$ is small

$$dh \approx du = CdT$$

Where C can be either constant volume and constant pressure specific heat

$$\text{If } C \neq f(T) \Rightarrow (h_2 - h_1) \approx (u_2 - u_1) \approx C(T_2 - T_1)$$

Internal energy , enthalpy and specific heats of solids and liquids

For solids and liquids $C_p \approx C_v \approx C$

It is found that C of a solid/ liquid depends on temperature only ,
i.e., $C = f(T)$

Internal Energy Change: $du = CdT$

$$u_2 - u_1 = \int_1^2 C(T) dT$$

For small temperature difference C can be taken as $C_{avg} = \frac{\int_1^2 C(T) dT}{T_2 - T_1}$

Enthalpy Change: $h = u + pv$

$$dh = du + pdv + vdp = du + vdp$$

For solids v is very small: $dh = du = CdT$ $h_2 - h_1 = C_{avg}(T_2 - T_1)$

For liquids two special cases exists:

1. Constant Pressure Process: Heater etc,

$$dp \rightarrow 0 \Rightarrow dh \approx du \approx cdT$$

2. Constant Temperature process: examples: Pumps

$$\Delta T = 0 \Rightarrow du = 0; dh = vdp$$